

1. SELECT DISTINCT department
FROM students;

department
IT
HR
Finance

2. SELECT department, AVG(age) AS avg-age
FROM department
GROUP BY department;

department	avg-age
IT	20.5
HR	22
Finance	23

3. SELECT department, COUNT(student-id) AS student-
count
FROM students
GROUP BY department
HAVING COUNT(student-id) > 1;

department	student-count
IT	2
HR	2

4. SELECT student-id, name, age, department
FROM students
WHERE age BETWEEN 21 AND 23;

student-id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. SELECT student_id, name, age, department
FROM students
WHERE department IN ('IT', 'HR') AND
age > 21;

student_id	name	age	department
2	Bob	22	HR
4	Diana	23	Finance
5	Eve	22	HR

6. SELECT department, SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING total_credits > 5;

department	total_credits
IT	11

7. SELECT course_id, course_name, department, credits
FROM courses
WHERE credits <> 4;

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

department	total-salary	total-bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12. SELECT department, AVG(salary) AS avg-salary
 FROM salaries
 GROUP BY department
 HAVING avg-salary > 55000

department	avg-salary
IT	61000
Finance	70000

13 SELECT *, (salary + bonus) AS total-compensation
 FROM salaries
 WHERE total-compensation > 60000;

employee_id	name	department	salary	bonus
1	Tom		65000	
3	Spike		76000	
4	Tyke		67500	

14 SELECT department, SUM(budget) AS total-budget
 AVG(budget) AS avg-budget
 FROM projects
 GROUP BY department
 HAVING AVG(budget) > 20000;

department	total-budget	avg-budget
IT	270000	135000
Finance	80000	80000

15 SELECT project-id, project-name, department, budget
FROM projects
WHERE department <> 'Marketing' AND budget
BETWEEN 50000 AND 120000;

project-id	project-name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
5	HR Portal	HR	50000